

Plastic pollution in mangrove ecosystems: impacts, mitigation actions, and policy responses

260 sources reviewed · 1993–2026 · 7 thematic areas · 6 geographic regions

Summary of evidence

Thematic findings

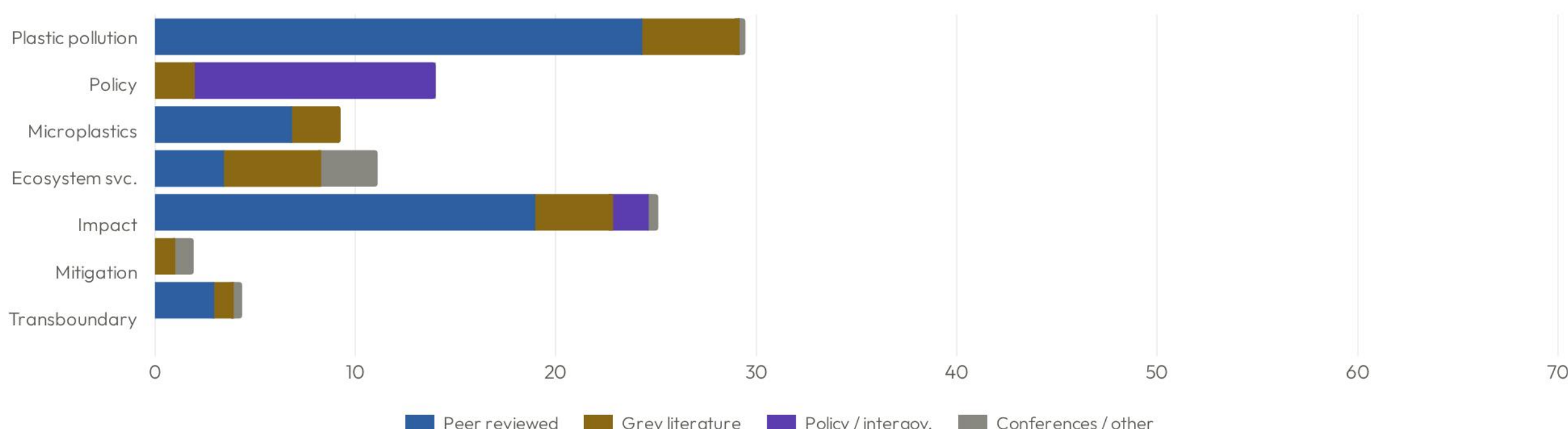
Evidence landscape

Source table

Assessing the quality and composition of the evidence base is essential for understanding how robust the review's findings are. This section examines literature types, quality scoring, and the intersection of research themes with geographic coverage.

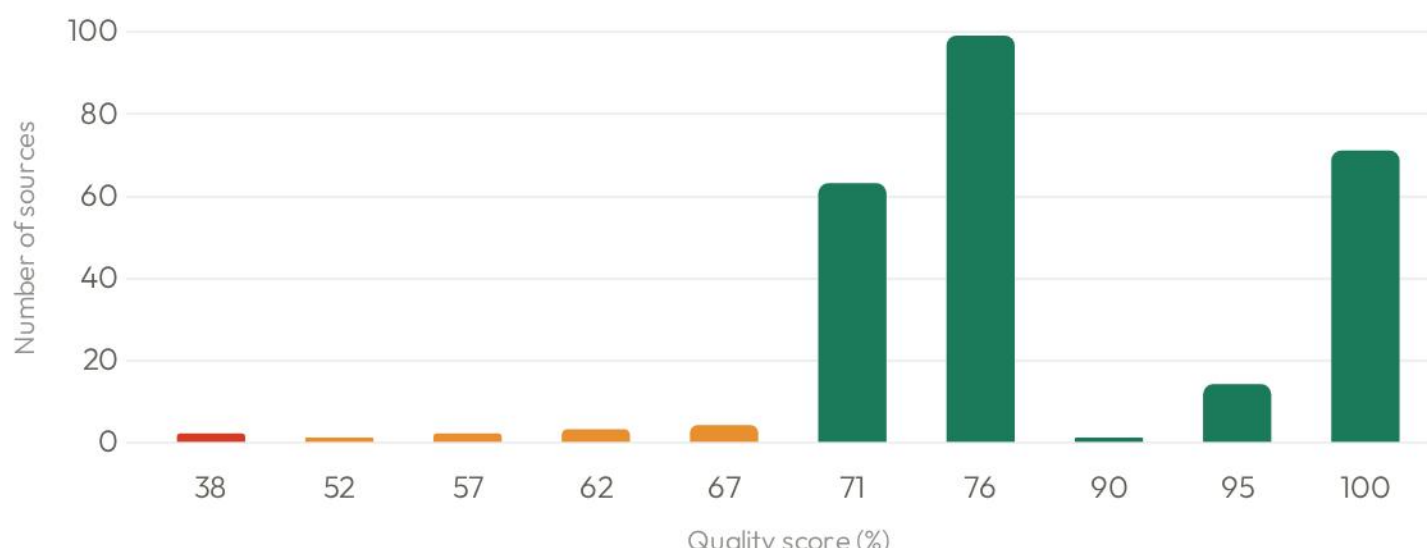
Source types across thematic areas

Peer-reviewed evidence is strongest in plastic pollution research (84%) and transboundary research (83%), but weakest in policy analysis (16%), which relies heavily on intergovernmental and legal sources.



Quality score distribution

Sources were scored on focus relevance (0–10), literature type (3–5), keyword match (0–3), and geographic specificity (0–3), giving a maximum of 21.



Quality bands

248 High (>70%)
95% of corpus

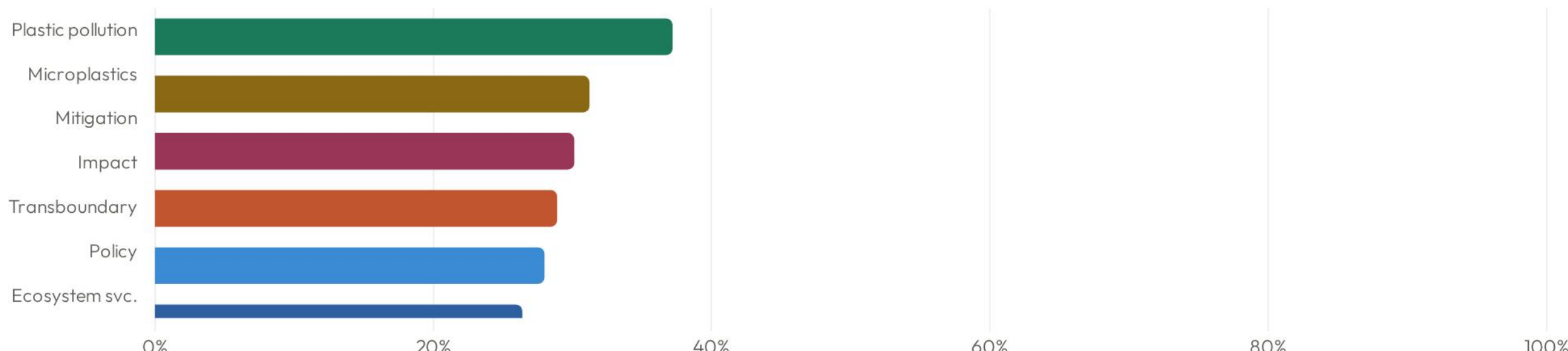
10 Medium (>40–≤70%)
4% of corpus

2 Low (≤40%)
1% of corpus

95% of scored sources fell in the high band (>70%); the lowest scores sit at 38%, indicating a well-curated corpus with strong minimum relevance. All 260 sources are now scored.

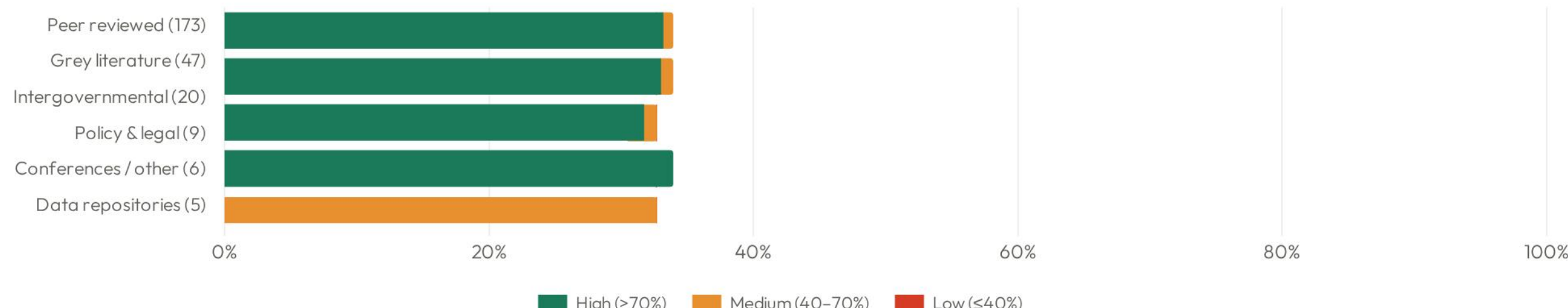
Evidence quality by thematic area

Mean quality score per theme. Plastic-pollution state assessments score highest (96%), reflecting on-target, peer-reviewed fieldwork; policy and ecosystem-services evidence scores lowest (~72%), where grey literature and broad-scope studies dominate. Lower-scoring themes are where evidence is thinnest and least mangrove-specific.



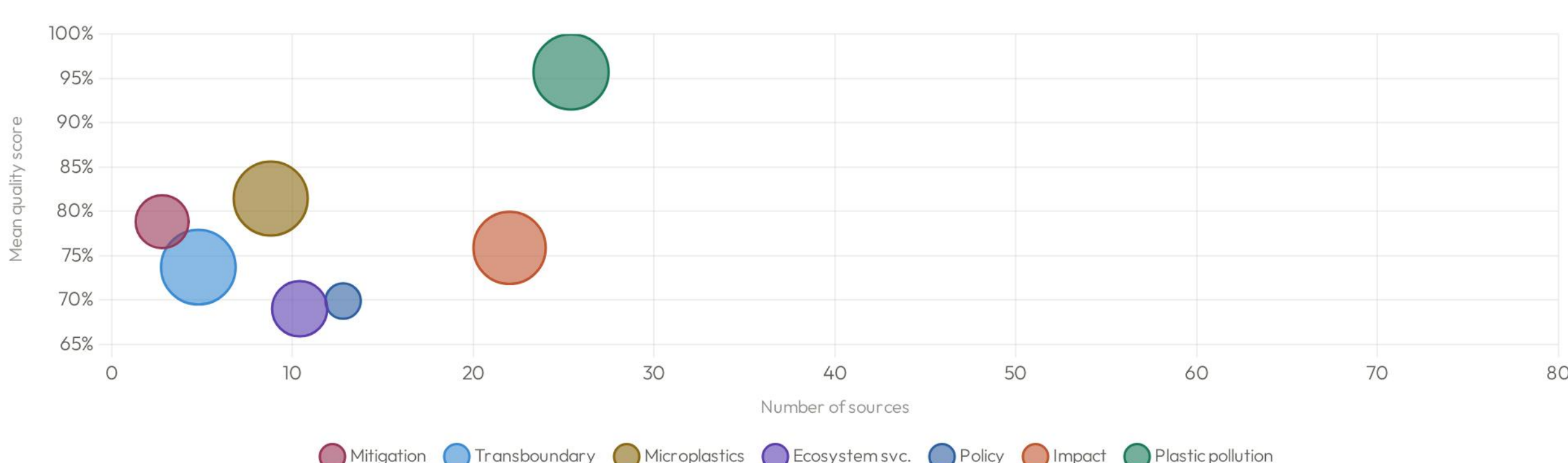
Quality by source type

How each literature type distributes across the quality bands — bars show composition, with source counts in brackets. Peer-reviewed, grey, intergovernmental and policy sources are almost entirely high-band; conference and newsletter material skews medium, and the only low-scoring items are two map/database records with minimal focus relevance.



Quantity vs quality by theme

Each bubble is a theme: horizontal position is the number of sources, vertical position the mean quality score, and bubble size the share that is peer-reviewed. Plastic-pollution research is large, high-scoring and heavily peer-reviewed; policy is sizeable but lower-scoring and mostly non-peer-reviewed; mitigation is small but scores well.



Research area × region heatmap

Cross-tabulation revealing where research effort concentrates and where significant gaps remain.

	Global	Asia/ ME	Africa	Europe	Americas	Pacific
Plastic pollution	34	28	3	–	4	1
Policy & governance	24	5	6	1	1	1
Microplastics & POPs	16	8	1	2	–	1
Ecosystem services	19	4	3	–	3	2
Impact assessment	39	7	1	3	8	1
Mitigation	4	5	2	1	1	–
Transboundary	12	2	1	1	–	1

The strongest evidence clusters are global-level impact assessment (39 sources) and mangrove plastic-pollution research (34 sources), followed by plastic-pollution fieldwork in Asia/Middle East (28 sources). Transboundary research is almost exclusively global-level. Mitigation evidence is most concentrated in Asia/Middle East and Global sources.

Dominant research keywords

The 14 most frequently tagged themes across all sources, reflecting the core conceptual territory of the field.

